

Module 14 – Generative AI Fundamentals

Learning Notes

Student Learning Summary

This document summarizes my understanding after studying the Generative AI Fundamentals module.

Overview

Generative AI creates new content such as text, code, images and more.

Unlike traditional AI that mainly classifies or predicts, Generative AI produces new outputs from patterns learned during training. It is widely used for chatbots, coding assistants, content creation and automation.

Difference between Traditional AI and Generative AI

Traditional AI focuses on prediction and classification. Generative AI creates new text, code, images and other content based on prompts.

History

Early chatbots appeared in the 1960s. GANs became popular in 2014. GPT-3 expanded language generation in 2020. ChatGPT made GenAI widely accessible in 2022. GitHub Copilot introduced AI-assisted coding.

Prompt Engineering

Prompt engineering means writing clear instructions for AI. Good prompts improve accuracy and save time.

Prompt Techniques

Zero-shot: ask directly without examples.

Example: Write a C# method to reverse a string.

Few-shot: provide examples before the task.

Chain-of-Thought: ask the model to solve step by step for complex problems.

Best Practices

Be specific, provide context, define the output format, review AI responses and improve prompts iteratively.

Ethics

Avoid biased prompts, verify AI-generated information, protect confidential data and use AI responsibly.

GitHub Copilot

GitHub Copilot is an AI coding assistant integrated into editors like Visual Studio Code. It suggests code, comments, tests and documentation.

Setup

Install the GitHub Copilot extension in VS Code, sign in with GitHub, activate Copilot and start typing comments or code.

Core Features

Code completion, function generation, boilerplate creation, documentation, refactoring, optimization and unit test generation.

Security

Always review generated code for bugs, vulnerabilities, licensing concerns and privacy issues before using it.

My Learning

I understood the basics of Generative AI, prompt engineering and GitHub Copilot. I learned that AI can improve developer productivity but every generated solution should be verified before production use.

Conclusion

Generative AI is becoming an important part of software development. Responsible use, good prompt writing and code review are essential for obtaining reliable results.